Fiscal Year:	FY 2019	**Task Last Updated:**	FY 02/19/2019
PI Name:	Motil, Brian J. Ph.D.		
Project Title:	Modeling and Analysis of Packed Bed Reactor Experimental Data from International Space Station		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	44135	**Congressional District:**	10
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2001 Fluid Physics 01-OBPR-02: Research and Flight Experiment Opportunities
Start Date:	03/01/2017	**End Date:**	10/15/2019
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
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Flight Program:	ISS		
Flight Assignment:	NOTE: Continuation per NSSC information of CoI's NNX14AD28G and NNX10AL37G to 80NSSC18K0813 with new title "Modeling and Analysis of Packed Bed Reactor Experimental Data from International Space Station" and new period of performance (Ed., 5/15/18, 4/28/21)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Balakotaiah , Vemuri Ph.D. (University of Houston (grants 80NSSC18K0813, NNX17AF91G))		
Grant/Contract No.:	Internal Project ; CoI grants 80NSSC18K0813, NNX17AF91G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: Continuation of "Modeling and Experimental Studies on Gas-Liquid Two-Phase Flow Through Packed Beds in Microgravity" internal project--Principal Investigator Dr. Brian Motil ; CoInvestigator Vemuri Balakotaiah, Ph.D., grant NNX14AD28G (and previously NNX10AL37G) to grant 80NSSC18K0813 with new title "Modeling and Analysis of Packed Bed Reactor Experimental Data from International Space Station" and new period of performance. Includes also information for project “GAS-LIQUID TWO-PHASE FLOW THROUGH PACKED BEDS IN MICROGRAVITY,” internal project--Principal Investigator Dr. Brian Motil ; CoInvestigator Vemuri Balakotaiah, Ph.D., grant NNX17AF91G (Ed., 5/15/18, 4/28/21) A basic understanding of two-phase flows through porous media is of interest in many chemical and biological processing systems as well as many geophysical applications. NASA technologies include adiabatic two-phase flows or two-phase flows with heat transfer that are relevant to life support systems, fuel cells, in-situ resource utilization, heat pipes, materials processing, production of pharmaceutical grade water, and transport of nutrients in soils. A very important unit operation that involves gas-liquid flow through porous media is the fixed Packed Bed Reactor (PBR). Examples of current space systems in which a fixed PBR is used include the Volatile Removal Assembly (VRA), the Integrated Advanced Water Recovery System (AWRS), and IntraVenous Water GENeration system (IVGEN). NASA is currently operating systems with packed bed reactors for ground and flight-based testing with an incomplete understanding of how the reduced gravity environment affects the performance and reliability of these reactors. This is especially critical when gas-liquid flows are involved. The expected outcome of this research effort is to develop a set of guidelines and tools to enable engineers to reliably design and operate packed bed reactors for microgravity as well as the lunar and Martian environments. Existing approaches for predicting the flow regimes and pressure drop for gas-liquid flow through a PBR are valid only for 1 g, and are based on liquid (or gas) holdup within the bed. Initial microgravity tests conducted in the reduced gravity aircraft and drop tower clearly show that the liquid holdup is significantly influenced by gravity. Furthermore, most of these models have only been verified using the "trickle flow" regime which does not exist in microgravity. Basic hydrodynamic models developed specifically for reduced gravity have been partially validated within the constraints of aircraft and ground facilities. However, typical flow ranges used for real systems require up to several minutes to develop fully which is far beyond the time scales available for ground tests. In addition, instabilities and hysteresis effects that may exist require long testing durations to establish new operating conditions. Thus, testing on the International Space Station (ISS) is required for complete validation of the semi-empirical models. Flight experiments can provide critical hydrodynamic information for a project with a broader scope that includes reduced gravity aircraft and ground-based (1 g) experiments. The main objective of the overall project is to develop and validate macroscopic equations that can be used in partial and microgravity conditions to accurately predict flow pattern transitions; pressure drops; and mass transfer rates in chemical and biological reactors in which gas-liquid flows occur through randomly packed beds. The hydrodynamic investigations addressed in the Packed Bed Reactor Experiment (PBRE) focus on the transitions between flow regimes (e.g., bubbly-to-pulse flow transition) and the associated pressure gradients over the range of relevant operating regimes of the PBRs (e.g., liquid and gas flow rates, particle sizes, and fluid viscosities). These design tools provide a fundamental understanding of these phenomena in low gravity.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The PBRE provides a fundamental investigation of two phase flow through porous media in the absence of gravity. This enables a direct study of the other forces that impact the flow and reaction rates such as surface tension, inertia, and viscous forces. These new insights will impact reactor designs on Earth.
Task Progress:	The past year has been focused on organizing, reducing, and evaluating the detailed data returned on hard drives to the Science Team. All raw data was organized and submitted to upload into the PSI (Physical Sciences Informatics) database in July, 2018, and the processed data (with calibration curves) was delivered in October, 2018. Science support was also provided to the engineering team to revise requirements for the re-flight in 2020 which included modifications to the mixing section; different packing sizes; and a revised test matrix. A draft manuscript has been prepared based on analysis of ISS data for possible publication in AIChE Journal (American Institute of Chemical Engineers). This manuscript also compares the ISS results to earlier models developed based on aircraft flights (published in 2003). The effects on pressure drop and phase distribution are discussed as well as the effect of a non-wetting packing material (Teflon). Experimental results to study the transient flows, i.e., sequentially increasing and decreasing the flow rate of either the gas or liquid phase while holding the other constant are discussed. Key findings include good agreement with the pressure drop models extended to much lower liquid Reynolds numbers as well as potentially new flow regimes. Interestingly, little or no hysteresis in pressure drop was observed during the transient flow tests. This is counter to observations published in a number of papers in normal gravity co-current gas-liquid down flows.
Bibliography Type:	Description: (Last Updated: 04/26/2024)
Articles in Peer-reviewed Journals	Salgi P, Krayem S. "Modeling pulse properties near the bubble-to-pulse transition in randomly packed beds." International Journal of Chemical Engineering. 2019 Jan;2019:Article ID 1635265. https://doi.org/10.1155/2019/1635265 , Jan-2019